
Credit Deterioration Early Warning System

From default prediction to risk monitoring and expected loss

This report summarizes the current credit risk model. It covers the data, model design, validation results, probability calibration, expected loss, and the emerging-risk watchlist. The final section lists the work still required before operational use.

Technical project report

14 September 2026

Technical summary

This project estimates default risk over 12-, 24-, and 36-month horizons. It also identifies loans whose risk increases over time and converts predicted default risk into expected loss. These outputs support three monitoring groups: Healthy, Watchlist, and High Risk. The model is a research prototype and is not ready for production decisions.

- **The baseline ranking is useful.** A logistic regression trained on loans issued through 2014 and evaluated on resolved 2015 loans achieved a ROC-AUC of 0.7295 and average precision of 0.4067. Because the test default rate was 0.202, average precision was about 2.01 times the random-ranking baseline.
- **Risk bands separate the portfolio clearly, but probabilities are too low in the riskiest bands.** For logistic regression, the observed default rate increased from 5.5% in the lowest-risk quintile to 42.3% in the highest-risk quintile. In that highest quintile, the mean predicted probability was only 34.4%, an underprediction of 7.9 percentage points.
- **Gradient boosting adds little discrimination.** Categorical-aware histogram gradient boosting increased ROC-AUC from 0.7295 to 0.7314 and average precision from 0.4067 to 0.4078. The gain is too small to justify the more complex model on performance alone. Both models still underpredict default in the highest-risk groups.
- **Longer-term loans carry disproportionate risk.** In the resolved 2015 test cohort, 60-month loans represented 34% of exposure but 53% of the modelled expected loss under a constant 60% loss-given-default assumption. Their observed default rate was 36.4%, compared with 14.9% for 36-month loans. This comparison must be read with a selection caveat: only 67.1% of all 2015 60-month loans had a resolved terminal status and entered the lifetime test.
- **Horizon-specific models provide the core early-warning view.** The 12-, 24-, and 36-month models achieved ROC-AUC values of 0.715, 0.701, and 0.698. Their ranking lift was strongest at 12 months, while the cumulative probability paths supported a small “emerging risk” group whose observed 36-month default rate was 24.0%, versus 13.4% under normal monitoring. A new discrete-time hazard benchmark keeps unresolved loans and produces monotone cumulative probabilities by construction, although its event timing remains approximate.
- **The watchlist economics are illustrative.** A review rule combines predicted loss, a 15% assumed intervention effect, a \$50 review cost, and 20% review capacity. It selects 84,149 loans and estimates \$40.5 million of net benefit under those assumptions. The estimate is not validated for deployment.

Bottom line. The model can rank loans for review and show when risk may emerge. Its probabilities are not accurate enough for pricing, capital, or accounting. Before production use, the project needs better event dates, post-origination borrower data, validation across multiple vintages, and approved economic assumptions.

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1 The models rank risk well, but understate its level

1.1 The lifetime model provides useful separation

The first modelling stage uses a simple and interpretable benchmark: logistic regression. Its task is not to make a hard yes-or-no prediction at a single threshold. Instead, it assigns every loan a probability of default, which can be used to rank loans, form risk bands, and estimate expected loss.

Table 1. Out-of-time performance on resolved loans issued in 2015.

Model	ROC-AUC	Average precision	AP / default-rate baseline
Logistic regression	0.7295	0.4067	2.01×
Categorical-aware boosting	0.7314	0.4078	2.02×

ROC-AUC answers a ranking question: if one defaulted loan and one non-defaulted loan are selected at random, how often does the model give the defaulted loan a higher score? The value 0.729 means that this ordering is correct about 72.9% of the time. Average precision focuses more directly on the rare positive class. A value of 0.406 against a 0.202 base rate indicates that the score concentrates defaults near the top of the ranking about twice as effectively as random order.

The highest-risk fifth of the test set defaulted at 42.3%, while the lowest-risk fifth defaulted at 5.5%. The highest-risk group therefore defaulted about 7.7 times as often. This shows that the score can separate loans for screening and review prioritization.

1.2 Calibration, rather than ranking, is the immediate weakness

Figure 1 compares the average predicted probability in each risk quintile with its observed default rate. Points on the diagonal are perfectly calibrated. Points above it mean that realized defaults exceeded the model's prediction.

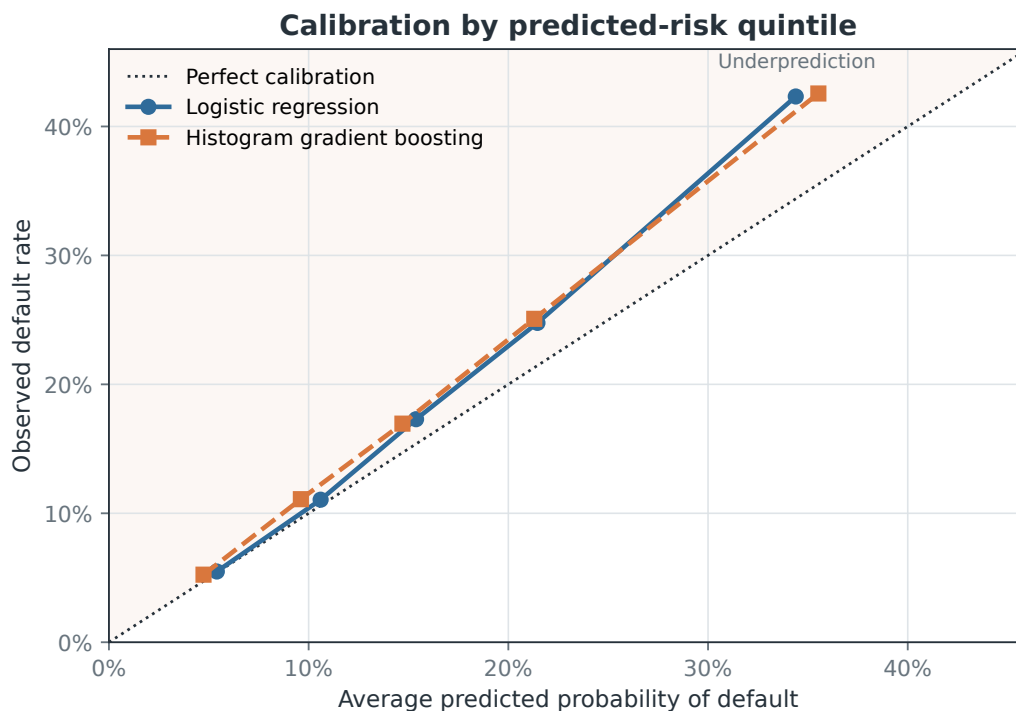


Figure 1. Calibration by predicted-risk quintile for the two lifetime models. Each point represents approximately one fifth of the 375,546-loan test set.

Both curves are close to the diagonal in the lowest-risk bands, then move above it as risk increases. Logistic regression predicts 34.4% in its highest band, where 42.3% is observed. Gradient boosting predicts 35.5%, where 42.6% is observed. The original nonlinear benchmark closes only 0.9 percentage points of the high-band gap.

The model ranks loans better than it estimates absolute default probability. It may support review prioritization, but its probabilities are too low for pricing or loss estimation. The default rate increased from 17.0% in training to 20.2% in the 2015 test cohort. This 3.16 percentage-point shift helps explain the underprediction and shows why recent calibration data is needed.

The corrected extension implements this temporal calibration design: the model is fitted through 2013, calibrated on 2014, and evaluated on 2015. Calibration does not change ranking, but it improves Brier score from 0.1474 to 0.1450 and moves mean predicted PD from 15.7% to 17.3%, closer to the observed 20.2%. The remaining gap, especially in the highest decile, shows that recalibration is helpful but not sufficient.

1.3 More flexible classification produces only a marginal gain

Histogram gradient boosting was introduced to capture nonlinear relationships and interactions that an additive log-odds model may miss. In the corrected comparison, nominal categories are explicitly marked as categorical rather than treated as a continuous numeric order. Even so, its improvement is only 0.0019 in ROC-AUC and 0.0010 in average precision. On this feature set, complexity alone does not change the conclusion.

The small performance difference shows that gradient boosting adds little value with the available origination variables. This does not show which variables cause default, and the notebook does not report coefficients, permutation importance, SHAP values, or individual-feature ablations. The broad ablation does show that lender-assigned fields alone reach ROC-AUC 0.710, borrower and loan fields alone reach 0.716, and the full model reaches 0.729. The full model gains information from both sources.

2 Term, grade, and cohort composition concentrate risk

2.1 Longer terms are riskier, but the resolved-loan sample is selective

Figure 2 shows that resolved 60-month loans default more often than resolved 36-month loans. Longer-term loans also remain exposed for more time, so they require separate monitoring.

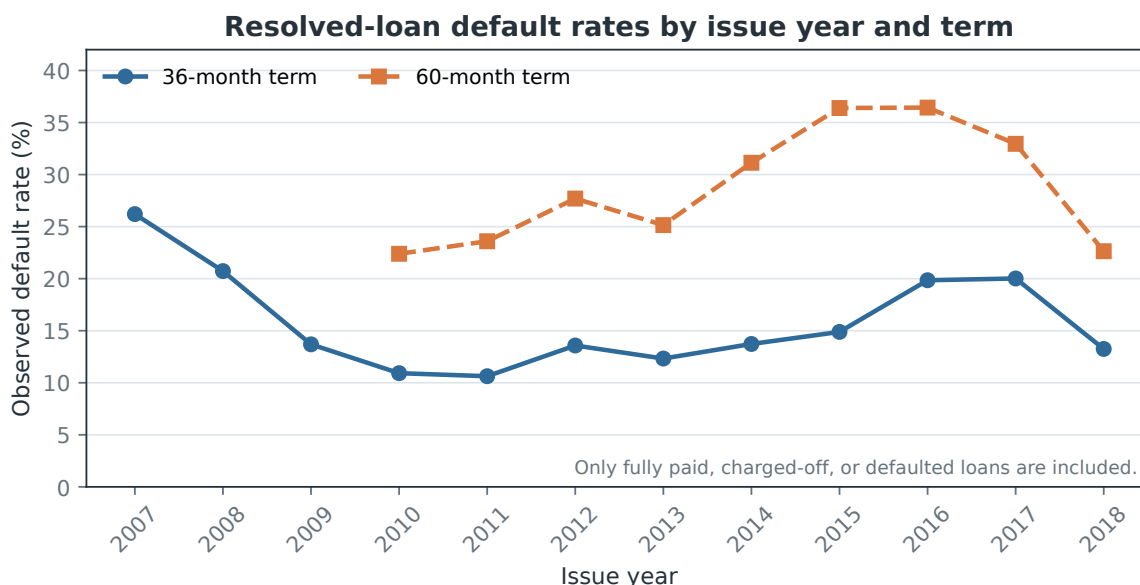


Figure 2. Resolved-loan default rates by issue year and contractual term. The chart describes loans with a terminal good or bad status; it is not a full vintage curve for unresolved loans.

For 2015, the lifetime test contains 283,026 resolved 36-month loans and 92,520 resolved 60-month loans. Within those samples, the model’s ranking is weaker than its overall ROC-AUC: 0.688 for 36-month loans and 0.669 for 60-month loans. Part of the aggregate performance therefore comes from separating the higher-risk term segment from the lower-risk one.

Table 2. 2015 resolved-loan results by term. Expected loss assumes LGD = 60% and EAD equal to original loan amount.

Term	Loans	Exposure	Predicted PD	Observed default	ROC-AUC	Avg. precision	Expected loss	EL rate
36 months	283,026	\$3.624bn	13.63%	14.89%	0.688	0.263	\$285.0m	7.86%
60 months	92,520	\$1.874bn	29.13%	36.39%	0.669	0.515	\$322.4m	17.20%

The 60-month segment has about half the exposure of the 36-month segment but slightly more modelled expected loss. This shows that loan term should be included in portfolio segmentation and monitoring.

However, this finding is also where sample construction matters most. Among all 421,095 loans issued in 2015, almost every 36-month loan had resolved by the data cutoff, but 43,227 of the 137,922 60-month loans were still current. Only 67.1% of the 60-month vintage entered the terminal-status test. The 36.4% figure is therefore a default rate among *resolved* 60-month loans, not among all loans issued on that term. It should not be used as an unbiased vintage default estimate.

2.2 Credit grade orders risk, while high grades remain underpredicted

LendingClub grades produce a strong monotonic risk ordering. Figure 3 shows that observed default increases from 5.5% in grade A to 54.2% in grade G. Under the report's illustrative loss assumptions, expected-loss rate rises from 3.5% to 25.4% of original exposure.

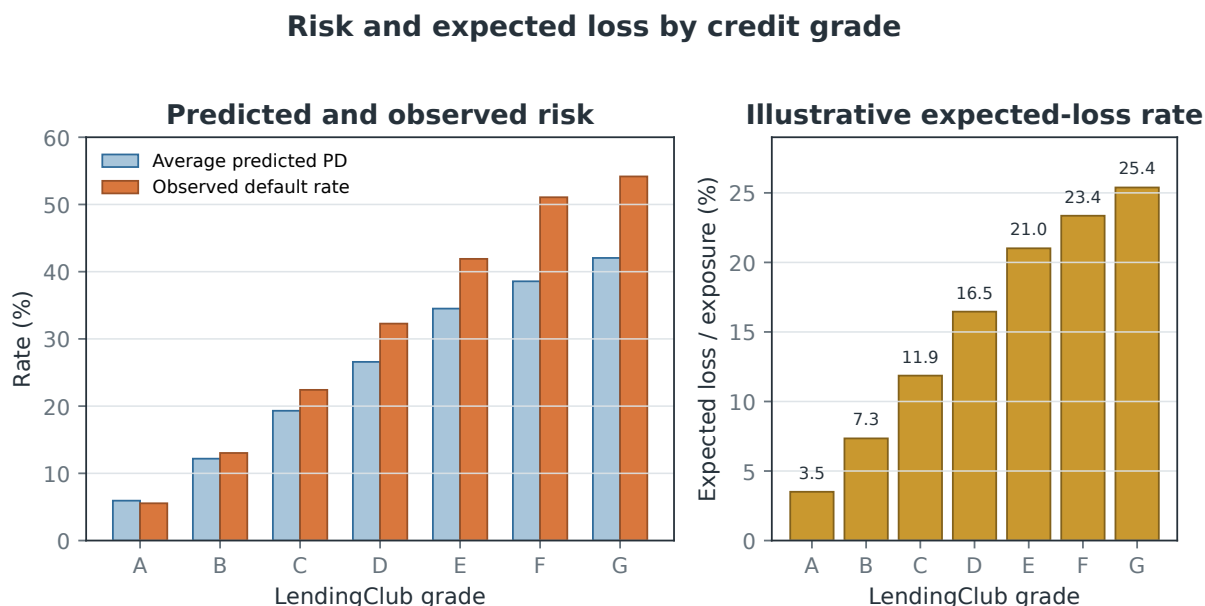


Figure 3. Predicted probability, observed default, and illustrative expected-loss rate by LendingClub grade for the resolved 2015 test cohort.

For grades D through G, observed default is increasingly higher than predicted default. This matters because review effort and expected losses are concentrated in these higher-risk grades.

Grade, sub-grade, and interest rate come from LendingClub's underwriting process. Including them supports portfolio monitoring but makes the model less independent as an underwriting tool. Borrower and loan fields alone produce ROC-AUC 0.716, compared with 0.710 for lender-assigned fields alone and 0.729 for the full model. This shows that both sources improve the full model.

3 Horizon models reveal when risk emerges

3.1 Near-term risk is easier to rank from origination information

Three models estimate cumulative default risk within 12, 24, and 36 months rather than whether a loan ever defaults. The 12-month model has the highest ROC-AUC and the greatest average-precision lift relative to its default-rate baseline.

Table 3. Out-of-time performance for horizon-specific models on 2015 loans with adequate outcome information for the relevant horizon.

Horizon	Test loans	Default rate	ROC-AUC	Avg. precision	AP lift
12 months	420,801	6.05%	0.7150	0.1364	2.26×
24 months	420,801	13.32%	0.7009	0.2563	1.92×
36 months	420,748	17.37%	0.6982	0.3166	1.82×

Average precision rises with the horizon because the positive class becomes more common. Its lift over the default-rate baseline moves in the opposite direction, from 2.26 times at 12 months to 1.82 times at 36 months. This is a more informative comparison across horizons: origination features distinguish early defaults most strongly, while later defaults are likely to depend increasingly on borrower behaviour, macroeconomic conditions, and other post-origination information.

3.2 Every horizon produces economically distinct risk bands

Figure 4 compares observed default rates across quintiles formed separately within each horizon. Within the 36-month model, observed default rises from 5.3% in the lowest quintile to 33.3% in the highest. The same ordering holds at 12 and 24 months.

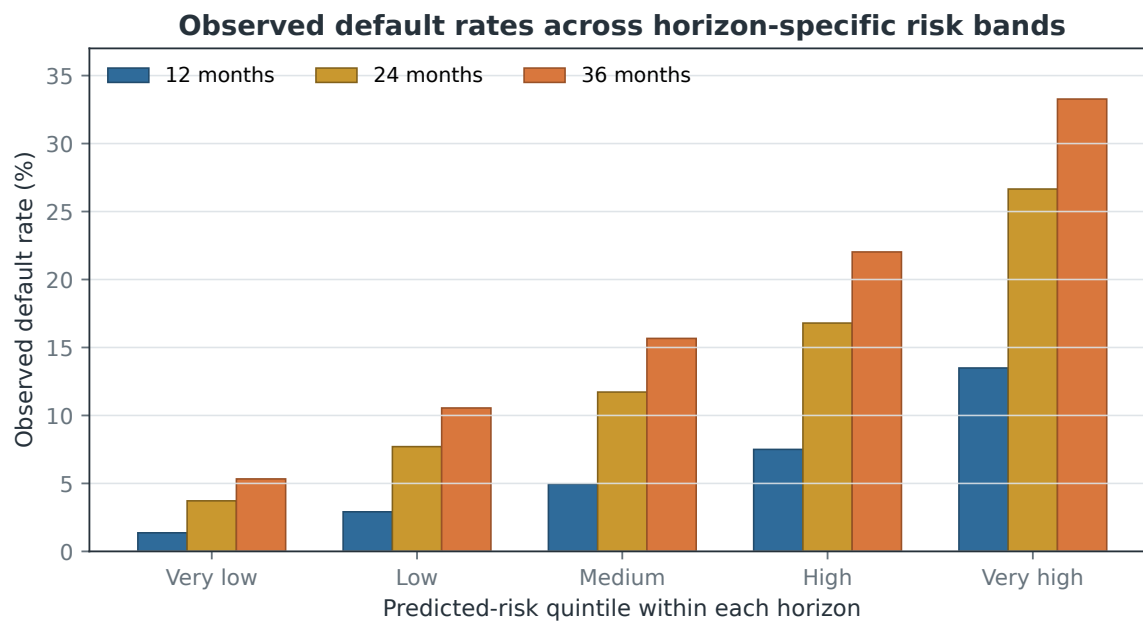


Figure 4. Observed cumulative default rates by horizon-specific predicted-risk quintile. Each band contains about 84,150 loans in the common 2015 cohort.

The horizon scores answer two separate questions: which loans are risky now, and which may become risky later? Near-term risk can trigger immediate review. Longer-term risk can determine monitoring frequency and early-warning alerts.

3.3 Risk acceleration identifies delayed deterioration

The notebook converts the three independent horizon probabilities into monotone cumulative paths. It then defines 36-month risk acceleration as adjusted 36-month PD minus adjusted 12-month PD. Figure 5 shows that loans with larger predicted increases also have markedly higher realized 36-month default.

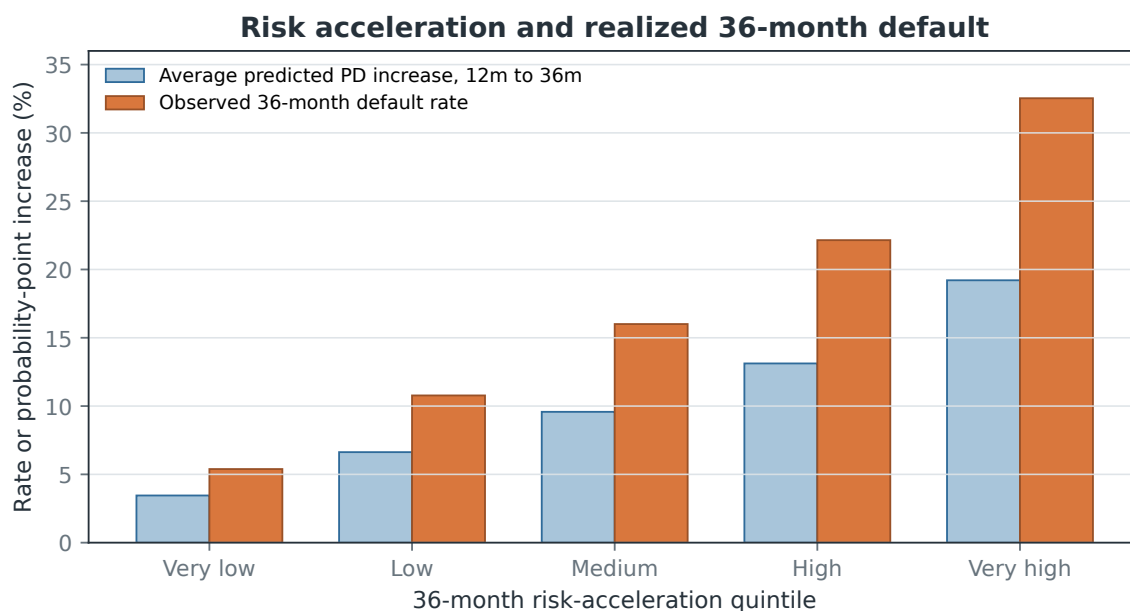


Figure 5. Average predicted increase from 12-month to 36-month PD and observed 36-month default rate by acceleration quintile.

Observed 36-month default rises from 5.4% in the lowest acceleration quintile to 32.5% in the highest. This shows that risk acceleration identifies loans with higher later default risk. It does not show what causes that deterioration.

4 A horizon-aware watchlist turns scores into an operating view

The final stage groups loans into three monitoring categories:

- **High immediate risk:** 12-month PD is at or above the 80th percentile.
- **Emerging risk:** 12-month PD is below the 60th percentile while the increase from 12-month to 36-month PD is at or above the 80th percentile.
- **Normal monitoring:** all remaining loans.

Table 4. Watchlist profile in the common 2015 horizon cohort. Expected loss uses adjusted 36-month PD, a constant 60% LGD, and original loan amount as EAD.

Group	Loans	Share	Avg. 12m PD	Avg. 36m PD	Actual 12m	Actual 36m	Exposure	36m expected loss
Emerging risk	3,132	0.74%	4.30%	21.11%	6.67%	23.98%	\$57.2m	\$7.3m
High immediate risk	84,150	20.00%	10.71%	28.40%	13.49%	32.67%	\$1.320bn	\$232.4m
Normal monitoring	333,466	79.26%	3.40%	11.89%	4.16%	13.45%	\$5.035bn	\$361.4m

The emerging-risk group is small, only 0.74% of the cohort, but its realized 36-month default rate is almost twice the normal-monitoring rate. Its 12-month default rate is much lower than the

high-immediate-risk group. This shows that the rule finds loans whose risk becomes material later rather than immediately.

Normal monitoring contains 79.3% of loans and about 60% of total modelled 36-month expected loss. This shows that the watchlist can prioritize reviews, but it cannot replace portfolio-wide loss measurement.

The three monitoring groups use percentile cutoffs, so the thresholds will change with the portfolio. The notebook also tests an economic rule based on expected loss avoided, review cost, and review capacity. Under the example assumptions, it selects 20% of loans. Those loans have a 29.6% observed 36-month default rate and \$2.005 billion of exposure. Before using this rule, the business must approve the review capacity, intervention effect, cost, and risk appetite assumptions.

5 What was modelled and which loans were eligible

5.1 Source population and terminal-status label

The source file, `accepted_2007_to_2018Q4.csv.gz`, contains LendingClub accepted-loan records from 2007 through 2018 Q4. After requiring non-missing loan status and loan amount, the working source contains 2,260,668 rows. The lifetime-default analysis then keeps only loans whose terminal status can be interpreted unambiguously as good or bad.

Table 5. Population flow for the lifetime-default model.

Stage	Loans	Definition
Source after required fields	2,260,668	Non-missing <code>loan_status</code> and <code>loan_amnt</code> .
Resolved outcome sample	1,348,099	Fully paid, charged off, or defaulted, including legacy policy-status variants.
Resolved defaults	269,360	Charged off, default, or legacy charged-off policy status.
Training cohort	453,809	Issue year no later than 2014; default rate 17.03%.
Test cohort	375,546	Issue year 2015; resolved-outcome default rate 20.19%.

The binary target is

$$y_i = \begin{cases} 1, & \text{if loan } i \text{ is charged off or defaulted,} \\ 0, & \text{if loan } i \text{ is fully paid.} \end{cases}$$

Loans marked current, late, or in grace period are excluded from this first stage. That choice gives a clean binary label but creates the resolved-outcome selection described earlier.

5.2 Origination features and preprocessing

The notebook selects 22 source fields, then converts issue date into year and month. Issue year is used to define the temporal split rather than as an intended predictor. The modelled information covers five broad areas:

Table 6. Feature groups used in the project.

Feature group	Variables
Loan structure	Loan amount, contractual term, interest rate, installment, and issue month.
Lender risk labels	Grade and sub-grade.
Borrower capacity	Annual income, employment length, home ownership, verification status, purpose, and debt-to-income ratio.
Credit history	Delinquencies in the past two years, FICO range, inquiries in the past six months, open accounts, public records, revolving balance, revolving utilization, and total accounts.
Time fields for horizon labels	Issue date and last payment date; these are used to form timing labels and are not intended as outcome predictors.

Missing numeric values are replaced with the training median. Missing categorical values are replaced with the most frequent training category. Logistic regression standardizes numeric variables and one-hot encodes categories. Employment length has the largest missing share at 5.83%; other selected fields are nearly complete, with revolving utilization missing in 0.067% and debt-to-income ratio in 0.028%.

The scikit-learn pipeline learns imputation and encoding from the training data and then applies them to the test data. This prevents test information from entering preprocessing.

5.3 Eligibility for the horizon targets

The horizon models require enough follow-up to decide whether a non-default has survived each horizon. The notebook uses last payment date as an approximate timing proxy and includes a row if a default appears within the horizon, if at least that many months are observed, or if the loan was fully paid earlier.

Table 7. Eligible observations and target rates for horizon-specific labels.

Horizon	All eligible rows	Overall default rate	Eligible 2015 test rows
12 months	1,901,296	6.01%	420,801
24 months	1,592,168	13.47%	420,801
36 months	1,422,033	18.18%	420,748

Longer horizons include fewer eligible loans because more follow-up is required. They also include more defaults because loans have more time to fail. Results must therefore be compared within each horizon’s eligible population.

6 How the modelling stages were built

6.1 Logistic regression as an interpretable baseline

For a feature vector x_i , logistic regression models the probability of default as

$$\widehat{\text{PD}}_i = \Pr(y_i = 1 \mid x_i) = \sigma(\beta_0 + x_i^\top \beta) = \frac{1}{1 + \exp[-(\beta_0 + x_i^\top \beta)]}.$$

The parameters are fitted by minimizing binary log loss with scikit-learn's default L_2 regularization:

$$\mathcal{L}(\beta) = - \sum_{i=1}^n \left[y_i \log(\widehat{\text{PD}}_i) + (1 - y_i) \log(1 - \widehat{\text{PD}}_i) \right] + \lambda \|\beta\|_2^2.$$

Logistic regression produces probabilities and has a transparent structure. It assumes effects are additive in log-odds. Without explicit interaction terms, it may miss combinations whose joint effect is stronger than the sum of their parts.

6.2 Histogram gradient boosting as the nonlinear comparison

The comparison model builds an additive sequence of shallow decision trees:

$$f_M(x) = f_0(x) + \sum_{m=1}^M \eta h_m(x), \quad \widehat{\text{PD}}(x) = \sigma(f_M(x)),$$

where h_m is the tree fitted at boosting step m and $\eta = 0.05$ is the learning rate. The notebook uses 200 iterations, at most 31 leaves per tree, at least 100 observations per leaf, L_2 regularization of 0.1, and random seed 42.

Numeric variables receive median imputation. Categorical variables receive most-frequent imputation followed by ordinal encoding, with unseen categories set to -1 . The corrected model passes an explicit categorical-feature mask to the learner, so those codes identify category membership rather than a continuous ordering. This preserves the computational advantage of the histogram learner without imposing an artificial scale on purpose, home ownership, and similar nominal fields.

6.3 Out-of-time validation

The main model comparison is trained on loans issued no later than 2014 and tested on loans issued in 2015. This is more realistic than a random split because it forces the model to score a later vintage. The calibration extension uses a stricter three-way split: fitting through 2013, probability calibration on 2014, and final evaluation on 2015. Both designs reveal temporal drift because the test cohort has a higher resolved-outcome default rate than the earlier cohorts.

The notebook does not tune hyperparameters through cross-validation. The results compare benchmark models rather than optimized candidates, and the uncertainty around the small metric differences is not quantified.

7 How performance, timing, and expected loss were measured

7.1 Discrimination and average precision

ROC-AUC can be written as the probability of correct pairwise ordering:

$$\text{AUC} = \Pr(\widehat{\text{PD}}_i > \widehat{\text{PD}}_j \mid y_i = 1, y_j = 0).$$

Average precision summarizes the precision-recall curve:

$$\text{AP} = \sum_n (R_n - R_{n-1}) P_n,$$

where P_n and R_n are precision and recall at successive score thresholds. For a random ranking, expected AP is approximately the positive-class prevalence. This makes $\text{AP}/\Pr(y = 1)$ a useful descriptive lift measure across targets with different default rates.

7.2 Calibration by risk band

For risk band k , the report compares the average predicted probability with the observed default rate:

$$\bar{p}_k = \frac{1}{n_k} \sum_{i \in k} \widehat{\text{PD}}_i, \quad \bar{y}_k = \frac{1}{n_k} \sum_{i \in k} y_i, \quad g_k = \bar{y}_k - \bar{p}_k.$$

A positive gap g_k means that the model underpredicts default in that band. This banded check is easy to communicate, although a production validation should also report Brier score, calibration intercept and slope, confidence intervals, and calibration within material subgroups.

7.3 Horizon labels and monotone probability paths

Let m_i denote the approximate number of months from issue date to last payment date. For horizon $h \in \{12, 24, 36\}$, the target is

$$y_i^{(h)} = \mathbb{1}\{\text{eventual default}_i = 1 \text{ and } m_i \leq h\}.$$

An observation is considered valid for horizon h when

$$V_i^{(h)} = \mathbb{1}\{y_i^{(h)} = 1 \text{ or } m_i \geq h \text{ or fully paid}_i = 1\}.$$

This rule removes non-defaults with insufficient follow-up unless they fully paid early. It is a pragmatic censoring rule, but it is not a formal survival model.

Because the three classifiers are fitted independently, their cumulative probabilities can occasionally violate monotonicity. The notebook applies

$$\tilde{p}_{12} = p_{12}, \quad \tilde{p}_{24} = \max(p_{12}, p_{24}), \quad \tilde{p}_{36} = \max(p_{12}, p_{24}, p_{36}).$$

Violations are rare: $p_{24} < p_{12}$ for 0.018% of the common 2015 cohort and $p_{36} < p_{24}$ for 0.576%. The 36-month acceleration statistic is then

$$\Delta_i^{36} = \tilde{p}_{36,i} - \tilde{p}_{12,i}.$$

The corrected extension also fits a discrete-time hazard benchmark over successive 12-month intervals. If $q_{i,t}$ is the conditional default probability in interval t for a loan still at risk, cumulative probability is

$$\Pr(T_i \leq h) = 1 - \prod_{t \leq h} (1 - q_{i,t}).$$

This construction makes cumulative PD monotone without post-processing and keeps unresolved loans until censoring. The benchmark is trained through 2013, calibrated on 2014 interval hazards, and evaluated on the common 2015 cohort. Its 12-, 24-, and 36-month ROC-AUC values are 0.676, 0.668, and 0.666. The benchmark ranks loans less accurately than the separate horizon models. It is a sensitivity check, not the preferred model.

7.4 Expected loss translation

The project converts probability into an illustrative credit-loss measure using the standard identity

$$EL_i = PD_i \times LGD_i \times EAD_i.$$

It assumes a constant $LGD = 60\%$ and sets exposure at default equal to the original loan amount. For the resolved 2015 lifetime cohort, this gives \$607.3 million of modelled expected loss on \$5.499 billion of original exposure, or 11.05%.

Expected loss puts the model output in monetary terms, but this calculation is simplified. Actual EAD changes with amortization and prepayment. LGD changes with recoveries, collateral, costs, and time. Accounting or capital use would also require discounting, scenario weights, and policy definitions.

8 What the results do not yet establish

The following limitations restrict how the results can be used.

1. **Resolved-outcome selection is the main limitation of the lifetime model.** Current and delinquent loans are excluded. This is especially important for 60-month loans issued in 2015, only 67.1% of which have a terminal status in the source snapshot. The resulting default rates are not unbiased vintage rates.
2. **Default timing is approximate.** The horizon labels use `last_pymnt_d`, not a clean charge-off or legal default date. Last payment may precede the actual default event, so the models should be described as approximate early-warning models.
3. **Calibration changes over time and by segment.** The later test vintage has a higher base rate, and underprediction is strongest in high-risk, long-term, and weak-grade groups. Any use of PD in pricing or expected loss needs a recent calibration sample and segment-level checks.
4. **Some predictors embed the lender's existing risk assessment.** Grade, sub-grade, and interest rate are assigned through LendingClub's underwriting. Their inclusion is reasonable for portfolio monitoring, but it can overstate the model's independent underwriting contribution. The completed ablation shows that borrower and loan fields alone retain useful discrimination (ROC-AUC 0.716), while the full model remains stronger (0.729).

5. **Model interpretability has not yet been demonstrated empirically.** Logistic regression is structurally interpretable, but the notebook does not report coefficients or stability. Grade, sub-grade, and interest rate are also highly related, so individual coefficients may be unstable even when prediction is sound.
6. **The corrected categorical treatment is only one robustness check.** The updated histogram model marks encoded nominal features as categorical. Alternative learners and encodings should still be compared under the same rolling temporal validation design.
7. **Metric uncertainty is not quantified.** The comparison uses one out-of-time year and reports point estimates without bootstrap confidence intervals, vintage-by-vintage backtests, or statistical tests of the small model difference.
8. **The event-time benchmark remains approximate.** The new discrete-time hazard model handles censoring and produces monotone cumulative PD, but it still uses last payment as a default-time proxy and treats prepayment as a competing event censored at payoff. A validated competing-risks design is still needed.
9. **Expected loss is illustrative.** Constant LGD and original loan amount as EAD are transparent assumptions, not validated recovery and exposure models. The results are not IFRS 9, CECL, regulatory-capital, or pricing estimates.
10. **External validity is limited by data vintage.** The file ends in 2018 Q4. Product policy, borrower mix, servicing, and economic conditions may differ materially in a current portfolio.

The raw population counts, status definitions, temporal cohort sizes, and horizon label rates in the notebook were independently reproduced from the supplied source file. Model-performance values in this report come from the corrected notebook's stored outputs. The notebook was executed end to end in a fresh kernel: all 71 non-empty code cells completed and no error outputs remained.

9 Working checklist for the next iteration

The following work is required before production use. The order matters because later modelling and operating decisions depend on reliable data and event definitions.

Phase 1: Establish the data and target foundation

- ☐ **Audit the available post-origination data.** Confirm whether monthly payment history, days past due, outstanding balance, utilization, bureau updates, hardship events, recoveries, and relevant macroeconomic series are available. Produce a data dictionary, coverage table, and explicit gap list before choosing a dynamic model.
- ☐ **Build a leakage-safe loan-month panel.** Create one observation per loan per reporting month, preserve only information known at that date, and add automated checks for duplicate periods, impossible dates, outcome leakage, and inconsistent balances. This panel becomes the canonical dataset for subsequent deterioration modelling.
- ☐ **Define deterioration and the event clock.** Agree on operational definitions for delinquency migration, material PD increase, Watchlist entry, High Risk entry, default, cure, prepayment,

and exit. Replace last payment with the best available event dates and state how competing events will be handled.

Phase 2: Develop a genuinely dynamic risk model

- **Create simple behavioural benchmarks first.** Compare the current origination model with transparent additions such as recent delinquency status, missed-payment counts, payment-to-installment ratio, balance trend, and utilization change. Require each added signal to improve out-of-time warning performance.
- **Fit a monthly event-time model.** Estimate conditional 3-, 6-, and 12-month default risk with a discrete-time hazard or competing-risks model using time-varying features. Compare it with the existing horizon classifiers on the same loan-month cohorts and cutoffs.
- **Model risk-state transitions.** Estimate movement among Healthy, Watchlist, High Risk, Default, Cure, and Exit. Report transition matrices, state persistence, and time from first warning to default rather than evaluating only terminal classification.

Phase 3: Turn scores into stable and explainable alerts

- **Set state rules with persistence and hysteresis.** Define entry, escalation, cure, and exit thresholds using calibrated absolute PD, change in PD, and severe-event overrides. Require persistence where appropriate so small score movements do not repeatedly move an account between states.
- **Measure whether warnings arrive early enough.** Report the share of defaults and loss exposure identified 3, 6, and 12 months in advance, median alert lead time, precision and recall at review capacity, false alerts, and average time spent on the Watchlist.
- **Explain each alert at borrower level.** Show current PD, change from the previous observation, horizon of concern, exposure, expected loss, recent behavioural events, and a small set of stable reason codes. Validate that the explanations remain consistent under correlated predictors.

Phase 4: Validate decisions and prepare for controlled use

- **Run rolling temporal and segment backtests.** Repeat fit–calibrate–test splits over multiple vintages. Track discrimination, calibration, loss capture, warning lead time, and state stability with confidence intervals by term, grade, and material borrower segment.
- **Develop realistic LGD and EAD components.** Estimate recoveries, collection costs, outstanding exposure, amortization, and prepayment so that the economic layer reflects loss at the warning horizon rather than original balance and constant severity.
- **Validate intervention economics.** Replace the illustrative 15% effectiveness, \$50 review cost, and 20% capacity with policy-owned estimates. Where possible, use a retrospective comparison or controlled pilot to test which actions reduce loss, for which accounts, and at what cost.

- **Define monitoring and governance before deployment.** Assign owners for data quality, model changes, thresholds, overrides, and alert outcomes. Set review triggers for drift, calibration failure, subgroup deterioration, capacity breaches, and realized loss, and document the conditions for rollback or model retirement.

10 Questions worth answering next

- How stable is the borrower-and-loan-only model’s discrimination after removing LendingClub grade, sub-grade, and interest rate across rolling vintages and borrower segments?
- Does the 2014 calibration improve later vintages and high-risk segments, or will calibration require a rolling or segment-specific design?
- Which features distinguish emerging-risk loans from loans that are risky immediately, after controlling for term and grade?
- How stable are the watchlist shares and default rates across issue months, vintages, and macroeconomic regimes?
- What review threshold maximizes expected loss avoided after review cost, false positives, and operational capacity are included?
- How much do realistic amortizing EAD and segment-specific LGD change the apparent concentration of loss by term, grade, and watchlist group?

A Detailed result tables

Table 8. Logistic-regression calibration by risk quintile.

Risk band	Loans	Mean predicted PD	Observed default	Gap
Very low	75,110	5.40%	5.49%	0.08 pp
Low	75,109	10.60%	11.05%	0.45 pp
Medium	75,109	15.38%	17.30%	1.92 pp
High	75,109	21.46%	24.77%	3.31 pp
Very high	75,109	34.39%	42.32%	7.93 pp

Table 9. Histogram-gradient-boosting calibration by risk quintile.

Risk band	Loans	Mean predicted PD	Observed default	Gap
Very low	75,110	4.74%	5.24%	0.50 pp
Low	75,109	9.61%	11.09%	1.48 pp
Medium	75,109	14.70%	16.95%	2.26 pp
High	75,109	21.30%	25.08%	3.78 pp
Very high	75,109	35.52%	42.56%	7.03 pp

B Reproducibility note

The report is based on the current corrected notebook `CreditRisk_fixed.ipynb` and the local LendingClub data extract `accepted_2007_to_2018Q4.csv.gz`. Figures in this document are re-drawn from the notebook’s executed numeric outputs so that the LaTeX source remains portable and the charts remain sharp when printed. The corrected notebook is the companion computational artifact for model fitting and row-level analysis; the original `CreditRisk.ipynb` remains unchanged.